Assignment no. 02

Title: Student Database using structure.

Objectives:

- To understand implementation of "Structure" as composite data structure.
- Understand the implementation of Student database and perform various operations on it usingStudent structure.

Problem Statement:

Write C/C++ program to store marks scored for first test of subject 'Data Structures and Algorithms' forN students. Compute

I. The average score of class
ii. Highest score and lowest score of class
iii. Marks scored by most of the students
iv. List of students who were absent for the test

Outcomes:

On completion of this assignment students will be able to-

- Implement the "Structure".
- Solve real world problem of creating student database and perform various operations on itlogically using Student Structure.

Software & Hardware requirements:

- Open Source C Programming tools like G++/GCC or Eclipse.
- 64-bit Open source Linux or its derivative.

Theory- Concept:

Structure:

What is Structure?

- A Structure is a collection of related data items, possibly of different types.
- A structure type in C/C++ is called struct.
- A struct is heterogeneous in that it can be composed of data of different types.
- In contrast, array is homogeneous since it can contain only data of the same type.
- Structures hold data that belong **together**.
- Examples:
    - Student record: student id, name, major, gender, start year, …

- Bank account: account number, name, currency, balance, …
- Address book: name, address, telephone number, …

- In database applications, structures are called records.
- Individual components of a struct type are called members (or fields).
- Members can be of different types (simple, array or struct).
- A struct is named as a whole while individual members are named using field identifiers.
- Complex data structures can be formed by defining arrays of structs.

Definition of a structure:

```
struct <structure-name>
{
        <data-type> <identifier_list>;
        <data-type> <identifier_list>;
        ...
} ;
```

Each identifier defines a member of the structure.

Example:

```
struct StudentRecord{

        char Name[15];
        int Id;
        char Dept[5];
        char Gender;
};
```

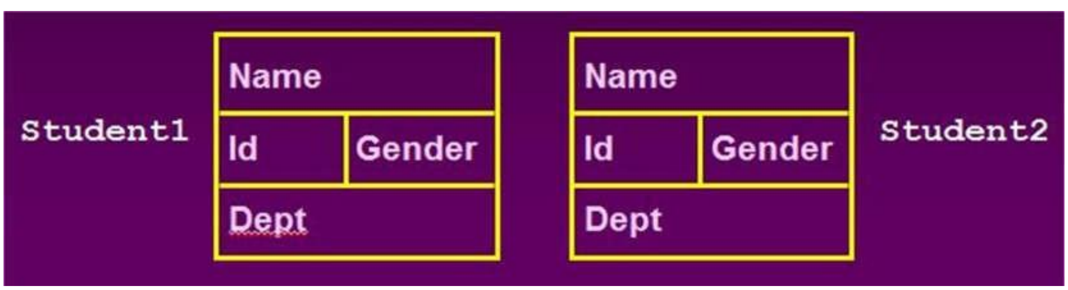
The “StudentRecord” structure has 4 members

Declaration of a variable of struct type:

```
struct <struct-name> <identifier_list>;
```

Example:

```
StudentRecord Student1, Student2;
```



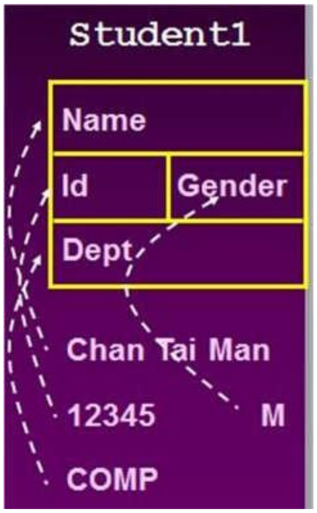
Student1 and Student2 are variables of StudentRecord type.

The members of a struct type variable are accessed with the dot (.) operator:

```
<struct-identifier-name>.<member_name>;
```

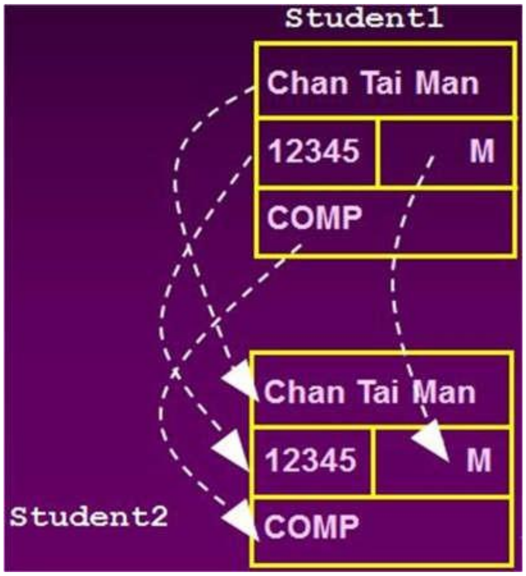
In above example the members of Student1 can be accessed as-

strcpy(Student1.name,"Chan Tai Man");
Student1.id=12345; Student1.Gender='M';
strcpy(Student1.Dept,"COMP");



- The values contained in one struct type variable can be assigned to another variable of thesame struct type.
    - Example:

Student2=student1;



Arrays of structures:

- An ordinary array: One type of data-
- An array of structs: Multiple types of data in each array element.

Arrays inside structures:

- We can use arrays inside structures.

Algorithm:

AlgorithmMaxnumber():

Step 1: Start

Step 2: Read: N inputs for DATA Array

Step 3: [Initialize] Set k = 1, LOC = 1 and MAX = DATA[1]

Step 4: [Increment Counter] Set K = K + 1 Step 5: if k > N, then:

Print: LOC, MAX and Exit if MAX < DATA[k] then:

Set: LOC = k and MAX = DATA[k] [End of inner If Structure]

[End of outer If Structure] Step 6: Go to Step 2.

Algorithm_Minnumber():

Step 1: Start

Step 2: Read: N inputs for DATA Array

Step 3: [Initialize] Set k = 1, LOC = 1 and MIN = DATA[1] Step 4: [Increment Counter] Set K = K + 1

Step 5: if k > N, then:

Print: LOC, MIN and Exit if MIN < DATA[k] then:

Set: LOC = k and MIN = DATA[k] [End of inner If Structure]

[End of outer If Structure] Step 6: Go to Step 2.

Flowchart:

Draw Your Own Flowchart Here!

Conclusion:

Hence we have learned Structure as a Data structure and how to use it to implement Student database.

Program Codes with sample output:

Assignment Questions (Write the Answer of following questions in Your Assignment book and Get it checked)

1. What is Union?
2. What is difference between c variable, array, structure and Union?
3. Write a program to compare two dates entered by user. Make a structure named Date to store the elements day, month and year to store the dates. If the dates are equal, display "Dates are equal" otherwise display "Dates are not equal".